Question:

Assume that B Corporation has the following total profit function equation: Total Profit = \pi = - $3,000 - $2,400 Q + $350Q^2 - $8.333Q^3 a) Compute the marginal profit b) Determine its output quantities

Answer Choices:

A. 22 and 7 units
B. 21 and 8 units
C. 23 and 6 units
D. 26 and 2 units
E. 24 and 4 units
F. 25 and 3 units
G. 20 and 5 units
H. 27 and 1 unit
I. 18 and 10 units
J. 19 and 9 units

Reasoning + Answer:

</tool_call> reasoning process here
The total profit function is given by:

ÏĢ = -3000 - 2400Q + 350QÂ² - 8.333QÂ³

a) Marginal profit (MP) is the derivative of total profit with respect to Q:

MP = dÏĢ/dQ = d/dQ [-3000 - 2400Q + 350QÂ² - 8.333QÂ³]

Differentiate term by term:

- d/dQ (-3000) = 0
- d/dQ (-2400Q) = -2400
- d/dQ (350QÂ²) = 700Q
- d/dQ (-8.333QÂ³) = -25QÂ² (since 3 ÃĹ 8.333 âīĪ 25)

So,

MP = -2400 + 700Q - 25QÂ²

b) To find the output quantity that maximizes profit, set marginal profit equal to zero:

-2400 + 700Q - 25QÂ² = 0

Multiply both sides by -1:

2400 - 700Q + 25QÂ² = 0

Divide entire equation by 25 to simplify:

96 - 28Q + QÂ² = 0

Rewriting:

QÂ² - 28Q + 96 = 0

Now solve the quadratic equation:

Q = [28 Â± âĪļ(28Â² - 4ÃĹ1ÃĹ96)] / 2
Q = [28 Â± âĪļ(784 - 384)] / 2
Q = [28 Â± âĪļ400] / 2
Q = [28 Â± 20] / 2

So,

Q = (28 + 20)/2 = 48/2 = 24
Q = (28 - 20)/2 = 8/2 = 4

Thus, the output quantities that maximize profit are Q = 24 and Q = 4.

Therefore, the correct answer is the pair (24 and 4).

<answer> E </answer>